Iván Marín Medrano

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Profile

Applied economist with eight years of experience in the research, public and private sectors; PhD in Applied Economics expected November 2026, trained in spatial econometrics, causal inference and network analysis under Julie Le Gallo and Jean Dubé. My research is on the causal evaluation of public intervention and the spatial structure of markets: what a policy actually changed, where the effect concentrates, and how long it lasts. I have estimated the effect of a public intervention on prices with difference-in-differences and reweighting, measured how competition decays with distance using geolocated data, and built indicators harmonised across millions of records. I work in R, Python, SQL and QGIS with reproducible workflows.

Education

Université de Bourgogne *2023 – Nov. 2026 (expected defense)*
PhD in Applied Economics France

↪ Thesis on empirical methods for competition, pricing and collusive behaviour in procurement and electricity markets
↪ Advisors: Julie Le Gallo (Institut Agro Dijon) and Jean Dubé (Université Laval)

Université de Bourgogne *2021 – 2023*
Master in Applied Economics France

Universidad Autónoma Metropolitana *2013 – 2017*
Bachelor in Economics Mexico

Research Experience

PhD Researcher | **CESAER, Université de Bourgogne, France** *Aug 2023 – present*

↪ **Policy evaluation**: measured the causal effect of a public inquiry on procurement prices in Quebec with **difference-in-differences**, panel event studies and **entropy balancing** on **20,153 bids**; prices fall by up to 11%, **more in low-competition markets**, and the effect fades over time.
↪ **Spatial economics**: built a Geographic Bidder Overlap index on about 2,850 contracts to measure **how competitive pressure decays with distance**; network models (ERGM) of interactions between firms. Published in *International Regional Science Review*.
↪ **Machine learning**: model to detect collusion and capacity withholding in the Italian wholesale electricity market, with Prof. Martin Huber (Université de Fribourg), published in *Energy Economics*.
↪ **Dissemination**: two peer-reviewed articles in 2026 and six presentations at international conferences, workshops and schools.

Visiting Researcher | **Université de Fribourg, Switzerland** *2026*

↪ Hosted by Prof. Martin Huber (causal inference, machine learning); the collaboration led to a publication in *Energy Economics*.

Data Scientist Assistant | **Government Transparency Institute, Hungary** *Nov 2025 – present*

↪ Developing and evaluating **machine-learning algorithms** to detect collusive bidding and procurement anomalies across **over 3 million contracts**, within a cartel screening project covering **9 European countries**.
↪ Engineering bid-level and market-level **screening indicators** from raw procurement records across countries.
↪ Co-authoring a working paper on cartel screening with the institute's research team.

Data Scientist Assistant | **Banque de France, France** *May 2022 – Aug 2022*

↪ Automated the processing and analysis of **international trade datasets**, contributing to research on firm dynamics.
↪ Replaced manual data preparation with **reusable scripts**, making results reproducible for the research team.
↪ Built a **Shiny app** and summary statistics that let economists explore the data without writing code.

Publications

- Marín, Iván, Julie Le Gallo, and Jean Dubé (2026). "The Spatial Structure of Competitive Interactions: Evidence from Public Procurement Networks in Quebec." *International Regional Science Review*. [link]
- Proz, Jeremy, Martin Huber, and Iván Marín (2026). "Machine Learning for Detecting Collusion and Capacity Withholding in Wholesale Electricity Markets." *Energy Economics*. [link]
- Hernández, José Luis, and Iván Marín (2022). "Welfare and Integration in Latin America: the Role of FDI." *Revista de Economía Mundial* 61: 91–110.
- Hernández, José Luis, and Iván Marín (2021). "Sustainable Financing of Public Investment: Evidence for Mexico, 1990–2019." *Revista Finanzas y Política Económica* 13(1): 43–75. [link]

Working Papers

- **Job Market Paper**: Marín, Iván. "Business as Usual: The Case of the Public Inquiry in Quebec, Canada."
- Fazekas, Mihály, Bence Tóth, Aly Abdou, and Iván Marín. "Public Procurement Cartels: A Multi-Country Screening Approach."

Seminars and Conferences

Spatial Econometrics Workshop, GATE Saint-Étienne (2025) · JMA, GAEL Grenoble (2026) · Economic Data Science Society, Banque de France (2026) · AFSE, Nantes (2026) · Winter School in Data Analytics and ML, Fribourg (2026) · AMASES Summer School on Networks, Rome (2023)

Skills

- **Policy evaluation**: difference-in-differences, event studies, entropy balancing and reweighting, heterogeneous treatment effects, panel data
- **Spatial analysis and mapping**: spatial econometrics, geolocated data, distance-decay effects, network analysis, thematic mapping (QGIS, R)
- **Data and software**: R, Python (pandas, polars, scikit-learn), SQL, Stata, Git, LaTeX; supervised machine learning
- **Languages**: Spanish (native), English (C1, scientific writing and presentation), French (C1)
- **Availability**: 1 December 2026

Other Professional Experience

Data Scientist | **Gestell, Mexico** *Oct 2020 – Sep 2021*

↪ Engineered behavioral features from historical transaction and repayment data to train an early-warning Random Forest classifier, reducing non-performing loans by 25%.
↪ Designed **end-to-end data pipelines** and curated datasets for experimental studies with **financial institutions**.
↪ Presented model results and recommendations to **non-technical** client stakeholders.

Economist | **Ministry of Finance, Mexico** *Jan 2019 – Sep 2020*

↪ Analyzed national and regional **macroeconomic indicators** to support fiscal policy decisions.
↪ Maintained time-series databases and prepared **briefing notes** and charts on the economic outlook for senior officials.

Data Analyst | **Coface, Mexico** *Jan 2017 – Dec 2018*

↪ Analyzed commercial activity data on companies across **Latin America** for a global trade-credit insurer.
↪ Consolidated data from several country sources into regional datasets and recurring **risk reports**.

References

- Julie Le Gallo, Professor, Institut Agro Dijon (CESAER), PhD advisor. julie.le-gallo@agrosupdijon.fr
- Jean Dubé, Professor, Université Laval, PhD advisor. Jean.Dube@esad.ulaval.ca
- Martin Huber, Professor, Université de Fribourg, co-author. martin.huber@unifr.ch